Worksheet 1

Probability Theory and Distributions

Master 1 Computer Science
Academic Year 2025–2026

Exercise 1 : Probability, Conditional Probability and Bayes' Theorem

A company develops an email filtering system.

Let

$$S = \{\text{the email is spam}\}$$

and

$$F = \{\text{the email is flagged by the filter}\}.$$

Assume that

$$P(S) = 0.25,$$

$$P(F|S) = 0.92,$$

and

$$P(F|S^c) = 0.08.$$

1. Calculate $P(F)$.
2. Calculate $P(S \cap F)$.
3. Calculate the probability that an email is spam knowing that it has been flagged by the filter.
4. Interpret the obtained result.

Exercise 2 : Discrete Random Variables

A machine produces electronic components. Each component is defective with probability

$$p = 0.03.$$

A random sample of 100 components is selected. Let X denote the number of defective components.

1. Determine the probability distribution of X.
2. Calculate the expectation and variance of X.

3. Calculate

$$P(X = 0).$$

4. Calculate

$$P(X \leq 2).$$

5. Can the distribution of X be approximated by a Poisson distribution? Justify your answer.

6. Using the appropriate approximation, estimate

$$P(X > 5).$$

Exercise 3 : Continuous Random Variables

The lifetime (in hours) of a sensor is normally distributed with mean

$$\mu = 1200$$

and standard deviation

$$\sigma = 150.$$

Let X denote the lifetime of a randomly selected sensor.

1. Determine the probability distribution of X.

2. Calculate

$$P(X < 1000).$$

3. Calculate

$$P(1000 < X < 1400).$$

4. Determine the lifetime exceeded by only 5% of the sensors.

5. Determine the interval centered at the mean containing approximately 95% of the observations.

Exercise 4 : Central Limit Theorem and Continuity Correction

The weight of cereal boxes produced by a factory has mean

$$\mu = 500 \text{ g}$$

and standard deviation

$$\sigma = 40 \text{ g}.$$

A random sample of

$$n = 64$$

boxes is selected.
Let

$$\overline{X}$$

denote the sample mean weight.

1. Using the Central Limit Theorem, determine the approximate distribution of $\overline{X}$.
2. Calculate

$$P(\overline{X} > 510).$$

3. Calculate

$$P(495 < \overline{X} < 505).$$

4. A box is considered underweight if its weight is less than 470 g.

   Assume that the number of underweight boxes in a shipment of 200 boxes follows a binomial distribution.

   Explain why a normal approximation may be used.

5. Using the continuity correction, estimate the probability that more than 20 boxes are underweight.

Exercise 5 : Reading Statistical Tables

Let

$$Z \sim \mathcal{N}(0, 1).$$

Using only the standard normal table, calculate:

1. $P(Z < 1.35)$
2. $P(Z > 1.96)$

3. $P(-1.5 < Z < 2)$

4. Find z such that

$$P(Z < z) = 0.95.$$

5. Find z such that

$$P(|Z| < z) = 0.99.$$